

subject to

$$\min_{\mathbf{x}} \mathbf{c}^T \mathbf{x}$$

$$Ax = b,$$

where

$$A \in \mathbb{R}^{2 \times 4}, \mathbf{b} \in \mathbb{R}^2, \mathbf{c} \in \mathbb{R}^4, \mathbf{x} \in \mathbb{R}^4,$$

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, c = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

We have a problem with $n = 4$ variables and $m = 2$ equality constraints. We consider the basic solution where x_1 and x_2 are in the basis:

$$\mathbf{B} = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}; \mathbf{B}^{-1} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}; \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}; \mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

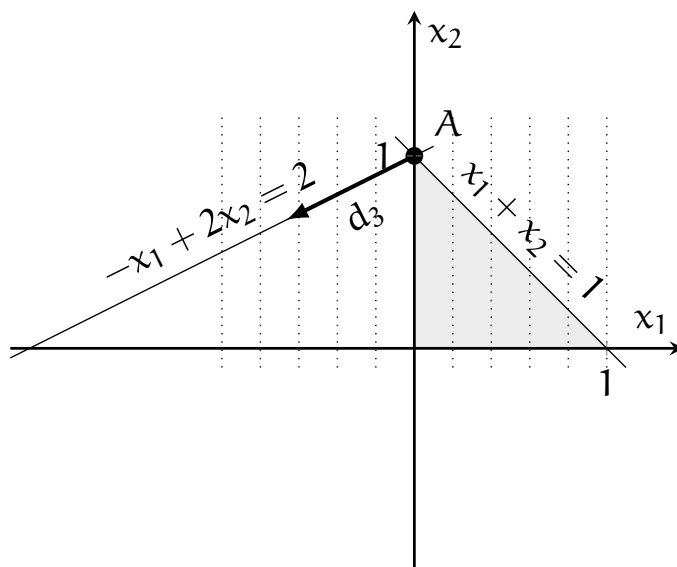
The basic entries of the basic direction corresponding to the non basic variable x_3 is calculated as

$$-B^{-1}A_3 = -\begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix}.$$

Therefore, the basic direction is

$$\mathbf{d}_3 = \begin{pmatrix} -\frac{2}{3} \\ -\frac{1}{3} \\ 1 \\ 0 \end{pmatrix}.$$

It is represented in the following figure.



The slope of the objective function is given by the directional derivative. As

$$c = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

we have

$$d_3^T c = -\frac{2}{3}.$$

It is negative. It means that the objective function is decreasing along that direction. The reduced cost is

$$\bar{c}_3 = c_3 - c_B^T B^{-1} A_3 = 0 - (1 \ 0) \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = -\frac{2}{3},$$

and we confirm that it is indeed the slope of the function in the basic direction.

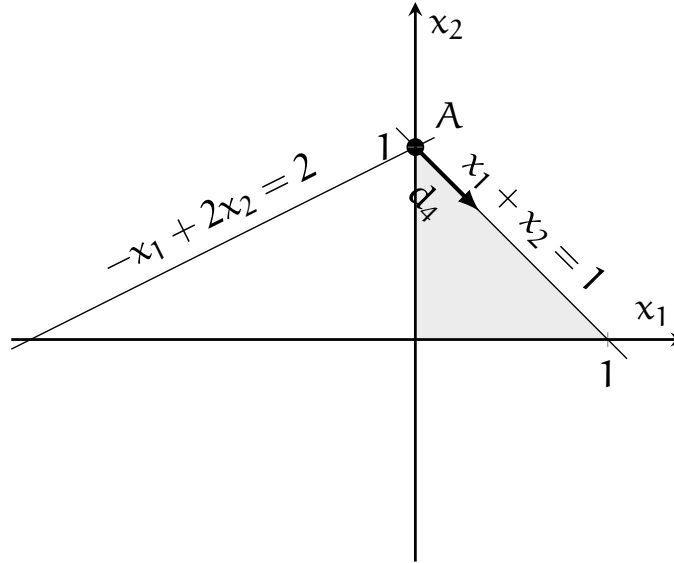
The basic entries of the basic direction corresponding to the non basic variable x_4 is calculated as

$$-B^{-1} A_4 = -\begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{3} \end{pmatrix}.$$

Therefore, the basic direction is

$$d_4 = \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{3} \\ 0 \\ 1 \end{pmatrix}.$$

It is represented in the following figure.



The slope of the objective function is given by the directional derivative. As

$$c = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

we have

$$d_4^T c = \frac{1}{3}.$$

It is positive. It means that the objective function is increasing along that direction. The reduced cost is

$$\bar{c}_4 = c_4 - c_B^T B^{-1} A_4 = 0 - (1 \ 0) \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{3},$$

and we confirm that it is indeed the slope of the function in the basic direction.

We have calculated the reduced costs for the two non basic variables. We now need to calculate them for the basic variables.

$$\bar{c}_1 = c_1 - c_B^T B^{-1} A_1 = 1 - (1 \ 0) \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0,$$

and

$$\bar{c}_2 = c_2 - c_B^T B^{-1} A_2 = 0 - (1 \ 0) \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 0.$$

This confirms the fact that the reduced costs of basic variables is always zero.